ATM-machine

Luka Huuhtanen TVTSPO25
Elias Lämsä TVTSPO25
Oskari Malkamäki TVTSPO25
Ukko Peltola TVTSPO25
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Introduction

The idea of this project was to develop a QT Bank ATM application that can be accessed through a RFID card and PIN code authentication. The system allows users to perform essential banking operations including withdrawals, deposits, balance inquiries, account switching, and viewing transaction history. Exchange rate information is retrieved through the Frankfurter API, enabling users to view live currency rates directly from the application..

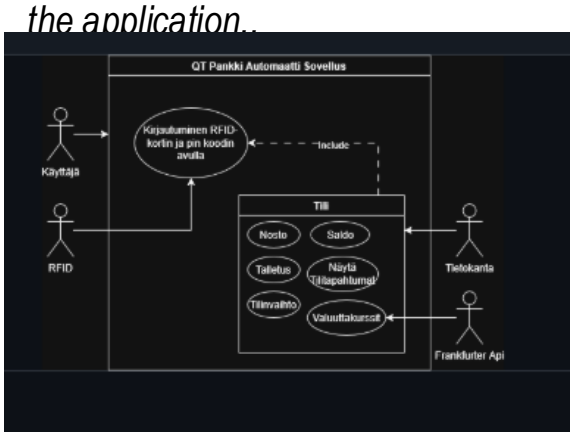


FIGURE 1. ATM-machine general fuctions

Objectives

The main objectives of this project were to implement RFID card and PIN code authentication for secure user login, to enable core banking operations such as withdrawal, deposit, balance inquiry, and account switching, to display transaction history and live currency exchange rates via the Frankfurter API, and to connect the application to a relational database for storing and managing banking data.

Methods

The project was developed using C++ and JavaScript as the primary programming languages alongside the QT Widgets framework to build the graphical user interface of the ATM application. The backend server was implemented using Node.js, which handled communication between the application and the database. JSON was used as the data format for exchanging information between the frontend and the backend. An RFID card reader was integrated as the hardware component responsible for reading user cards and initiating the authentication process together with PIN code verification

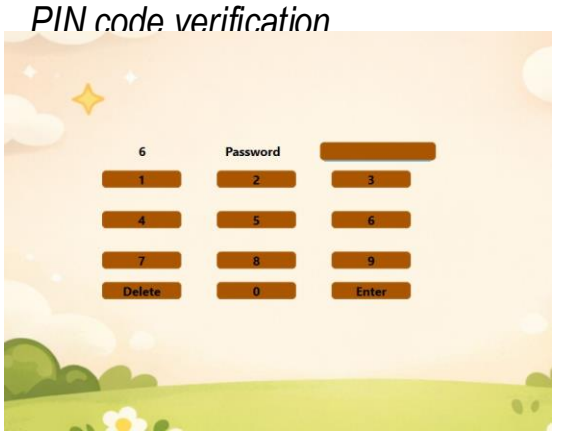


FIGURE 2. ATM-machine UI

Results

The project resulted in a fully functional ATM application where all planned features worked as intended, including RFID authentication, banking operations, and live currency exchange rates..

The login function sends the card number and PIN to the server via a POST request. Depending on the response status, it either clears the PIN for an invalid attempt, locks the card after too many failures, or on success extracts an authentication token from the JSON response and navigates to the main account screen.



FIGURE 3. QT app login

Conclusions

In conclusion, the project successfully resulted in a functional QT Bank ATM application that met all of its planned objectives. The combination of C++, QT Widgets, Node.js, and an RFID card reader produced a secure and working system capable of handling real banking operations. The project provided valuable experience in full-stack development, hardware integration, and database management.

References

1. *URL: https://github.com/tvt25spo-bank/group_11*
2. *URL: https://ohjelmistokehityksen-sovellusprojekti.oulu.net/*

Software Development Application Project

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